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Request for Information (RFI)

In response to: NSF_FRDOC_0001, Request for Information on the Development of an Artificial Intelligence (AI) Action Plan

Submitted to

Office of Science and Technology Policy (OSTP)
National Science Foundation
Networking and Information Technology Research and Development (NITRD) National
Coordination Office (NCO)

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GE Vernova is pleased to submit this response to the Request for Information, which advances the President's recent Executive Order 14179 on Artificial Intelligence. Energy is foundational to the President's plan and GE Vernova is uniquely positioned to serve this critical need.

Arguably, no single company is more invested, committed, and important to powering the needs of AI identified by the President than GE Vernova. GE Vernova is America's leading energy manufacturing company and has pioneered technologies that have spurred world-transforming changes and improved the lives of billions. Our technology helps produce 25% of the world's electricity, our power generation equipment is deployed in more than 140 countries, equips 90% of transmission utilities worldwide, and 40% of the world's electricity flows via GE Vernova software. GE Vernova is actively involved in all segments of the energy sector.

We thank the OSTP and NSF for the opportunity to submit comments on one of the most pressing policy issues of our time and for the critical leadership in this space. As described below, GE Vernova is uniquely positioned as a "1 of 1" American company to support the Administration's and the nation's success on these must-win goals.

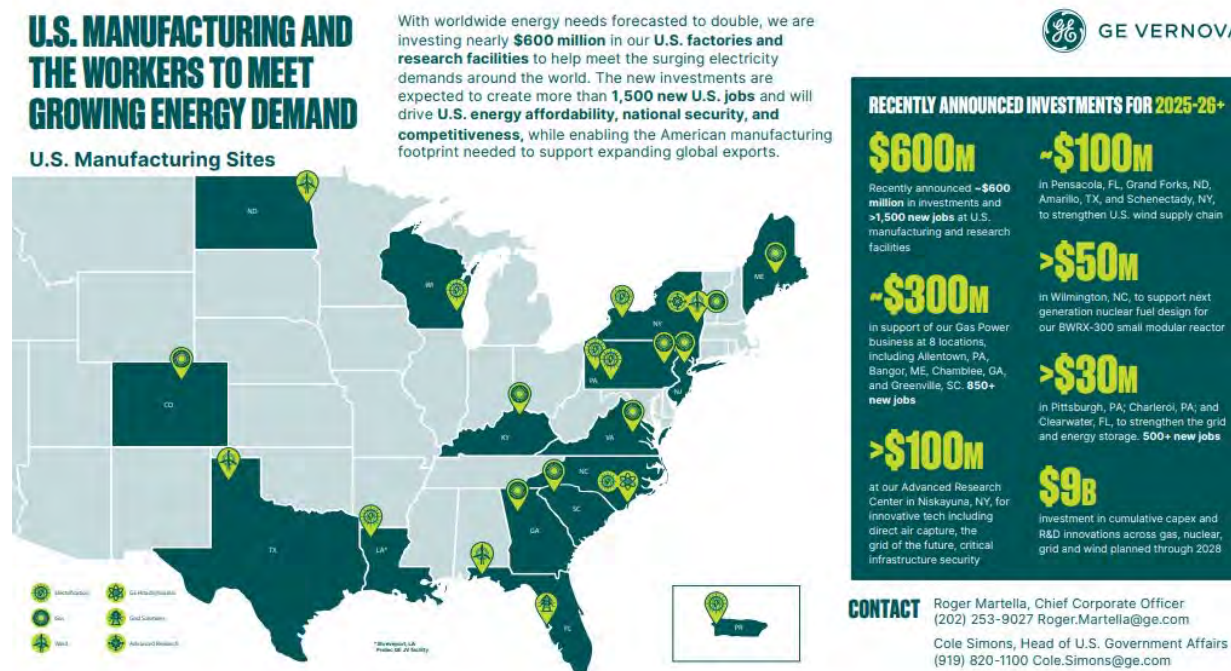
GE VERNOVA IS AMERICA'S ENERGY MANUFACTURER

GE Vernova is the only U.S. headquartered energy manufacturer of advanced technology and cutting-edge equipment across all energy sources and the grid – offering gas, nuclear,



wind, steam, and hydro power technology; as well as grid resiliency equipment, solutions, software, solar and storage. GE Vernova leads grid modernization and resilience efforts with a multi-layered approach to cybersecurity and reliability for the world's critical power systems. We service the products we sell and offer equipment upgrades that increase our products' efficiency and availability. We invest roughly \$1 billion per year into research to stay at the forefront of innovation and international competition. This innovation includes the only U.S. small modular reactor (SMR) currently under development, the BWRX-300, carbon capture and direct air capture, hydrogen integration into gas turbines, power conversion and storage solutions, and grid equipment and software designed to increase the reliability and efficiency of the grid.

GE Vernova employs over 18,000 workers in the U.S. This includes workers in all 50 states and Puerto Rico with 18 major factories across the country and more than 70 domestic facilities. **In February, we announced a new \$600 million investment into the expansion of our U.S. manufacturing base** covering gas, electrification systems, nuclear, and onshore wind, as well as our Advanced Research Center (ARC). The investment will create more than 1,500 U.S. jobs and is the first part of an expected \$9 billion global expenditure planned over the next four years.



GE VERNOVA IS UNIQUELY POSITIONED TO POWER U.S. ARTIFICIAL INTELLIGENCE (AI) LEADERSHIP

As energy and electricity has become recognized as a key enabler within the AI and data center race, **GE Vernova has emerged as the leading company for both grid connected and behind-the-meter projects powering AI and data centers.** In the past two months, GE Vernova has announced multiple partnerships to address the electricity component of the race, including:

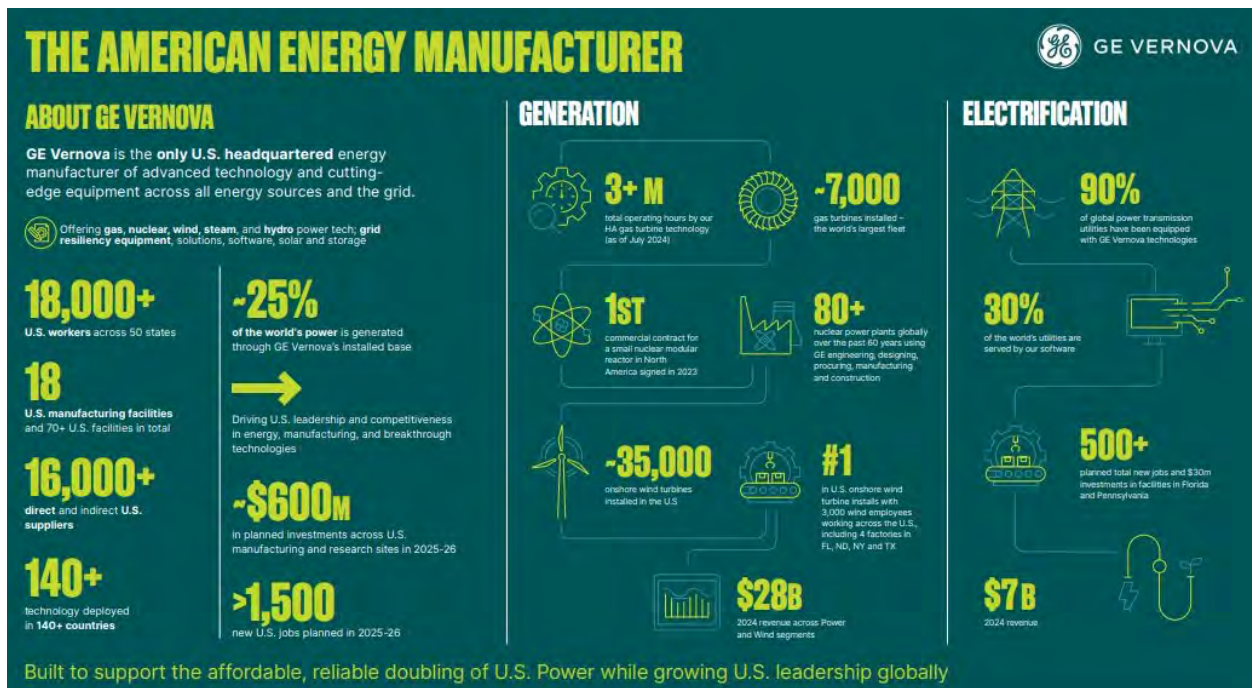
1. A partnership with Chevron and Engine No. 1 to develop scalable, reliable, multi-gigawatt power solutions for U.S. based data centers running on U.S. natural gas.



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2. A venture between GE Vernova, NRG Energy, and Kiewit Corporation to advance four projects totaling over 5 gigawatts (GW) of new natural gas combined cycle power plants for the ERCOT and PJM markets.
3. A strategic framework agreement (SFA) with Amazon Web Services (AWS) to support AWS's data center scaling with grid connectivity, and collaboration to address increasing energy demand, grid security and reliability, and decarbonized electric power systems.

In addition to the commercial role GE Vernova serves, GE Vernova has taken a leadership role in organizing industry discussion on AI and electricity. Recently, we hosted the GENERATE conference to bring together leaders from the Power and AI industry to accelerate cross industry discussions. We saw an opportunity to converge a multitude of conversations happening across the industry into meaningful roundtables to break down industry siloes. In our inaugural event, we had senior leaders from the key constituencies including hyperscalers, utilities, system operators, regulators, and engineering, procurement, construction (EPC) companies. As a convener, we made progress in ensuring a comprehensive look at the needs to be solved within the electricity and AI nexus. We have continued to convene discussions with the most significant parties in the ecosystem, including at CERAWEEK.



On the manufacturing side, GE Vernova is well positioned to provide the needed generation and electrification technology. **Our H-class gas turbine built in Greenville, SC**, recently surpassed more than 3 million commercial operating hours and is only a part of our gas turbine portfolio – with over 7,000 installed gas turbines globally. Additionally, during our history, GE Vernova has deployed over 80 nuclear reactors. Developed at the headquarters of our nuclear business in Wilmington, NC, **our next generation BWRX-300 SMR is poised to be the first SMR constructed by a U.S. company**. Already in development, the first deployment will be in Ontario, Canada, and operational in 2029. We also serve the wind market as the top ranked U.S. onshore wind turbine installer with over 35,000 onshore wind turbines installed in the U.S. and factories in Florida, North Dakota, New York, and Texas.



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In our **electrification business** we offer a multitude of products for electric grids, battery energy storage, and advanced power conversion. This includes but is not limited to transformers, switchgears, generator circuit breakers, power quality products, inverters, High Voltage Direct Current (HVDC) systems, power electronics systems, battery energy storage and grid management softwares. With facilities in Florida, Pennsylvania, Louisiana, Wisconsin, and North Carolina, over 90% of global transmission utilities have been equipped with GE Vernova technologies and 30% of the world's utilities utilize our software.

Because of GE Vernova's role and scale, we are uniquely well versed in the structure of the electricity system, AI and data center needs, and the interaction between the two. Our expertise allows us to evaluate the layered electricity system –at the local, state, Independent Power Purchasers (IPPs), utilities, Regional Transmission Organizations (RTOs) and Independent System Operators (ISOs), and federal levels – to provide insight into the opportunities and bottlenecks that need to be addressed to unlock the development of AI in the United States.

In this response, we offer our expertise to evaluate:

- Expected energy demand growth;
- A roadmap for serving electricity demand in an affordable and reliable manner; and
- Solving, via policy action and private sector innovation, expected bottlenecks associated with the rapid growth of electricity demand, in the areas of **dispatchable power, grid readiness, behind the meter systems, and innovation.**

I. ENERGY DEMAND GROWTH

Rapid proliferation of data centers and growth of AI are ushering in a new era of technological advancement and competitive advantage for the United States – as well as a global competitive risk if the U.S. does not foster an environment to maintain its advantage. **In the race to AI leadership, access to power generation is a key enabler and differentiator.**

Within 2-3 years, hyperscale data centers – critical to U.S. competitive advantage in AI – are likely to reach **2,000 MW per site**, consuming 10 million MWh of electricity annually. This is the equivalent of 1.7 million U.S. homes. Over the next 5-10 years, multiple hyperscalers are discussing even larger projects which would involve scaling to 5,000 MW at a site.

As data centers see escalating volumes of digital information and intensity in computations, electricity generation and grid capacity must surge to meet their needs. Supplying new generation and grid capacity to meet growing demand with reliable, uninterrupted power is vital to safeguarding the nation's economic stability, competitiveness, societal resilience, and security.

U.S. utilities expect grid connected electricity demand for data centers to grow **from 185 million MWh, or 4.5% of U.S. electricity demand in 2023, to 600 million MWh, or 12% of electricity demand by 2035.** This growth in electricity demand from data centers alone is the equivalent of more than three times the current electricity generation of New York State and would require more than 50,000 MW of dispatchable electricity generating capacity by 2035.

In addition to this strong electricity demand growth to support data centers and AI growth, there is nearly **200 TWh/y of additional industrial demand growth** through 2035 required to support



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the current administration's efforts to revitalize U.S. manufacturing, attracting investment in chip manufacturing, and onshoring of other large industrial facilities.

The large industrial and data center demand is heavily concentrated in certain geographies. Nearly **one-third of the combined growth is concentrated in PJM**, a regional transmission organization (RTO) that coordinates wholesale electricity markets across 13 states and the District of Columbia. The system operator estimates 40,000 MW of peak demand growth while retiring 40,000 MW of existing capacity in their region over the next 15 years. The Electric Reliability Council of Texas, **ERCOT, accounts for more than one-quarter of the combined industrial and data center demand growth**. The Mid-continent Independent System Operator, **MISO, includes nearly 15% of the incremental industrial and data center demand growth**.

The firm power needs of industrial and data center demand will require dispatchable generation as a foundation. U.S.-produced gas will need to play a significant role meeting the immediate power needs. Gas turbine capacity can come online quickly at scale, with emissions that can reduce the carbon intensity of the grid while meeting the growing demand. Nuclear SMRs together with renewables can be integrated into the mix in future years.

II. ROADMAP FOR MEETING U.S. DATA CENTER DEMAND

Meeting the surging power demands of data centers will require a dual-track solution over the next 10-years. Alongside upgrading existing U.S. generation equipment to produce an additional 16,000 MW across gas and nuclear, **adding 48,000 MW of dispatchable, flexible gas turbine capacity to quickly meet the surging need for dispatchable power and 22,000MW of generation from nuclear SMRs and renewables** is an achievable scenario. The approach supports a diversified power system capable of utilizing all energy resources to fulfill electricity need in a timely and cost-effective manner. New gas capacity additions would be front-end loaded to scale rapidly, while providing the dispatchable foundation required for reliability. This approach would also support some portions of capacity being added behind-the-meter.

As we build dispatchable capacity now, **gas remains the strongest option to support rapid growth with reliable, dispatchable generation with a small physical footprint** – while ensuring the U.S. continues to produce electricity with the lowest emission intensity and cleanest water and air in the world. **Over time, a greater proportion of renewables and SMRs could be layered in.** Diversity of supply, inclusive of renewables and nuclear, has added value helping certain customers achieve their stated broader objectives for “all of the above” power to meet affordability, reliability, and sustainability goals while reducing fuel price uncertainty risks. The approach requires expediting nuclear technologies available today **like GE Vernova's BWRX-300, the first SMR to be commercialized this decade**. Utilizing renewable technologies enhances the speed in which electricity demand can be met and recognizes that some areas may be attractive for renewables based on land availability, gas pipeline capacity, and a potential lack of carbon sequestration potential – thus reinforcing an important need for a US manufacturing base for renewables technologies. Over time, the approach enables emissions to be reduced in line with goals the utilities and hyperscalers have placed on emissions profiles.

To meet growing demand Figure 1 illustrates the proposed incremental electricity **generation** (in MWh/y) that new cumulative capacity could provide overlaid against data center demand growth. The figure is based on capacity factors and additional requirements for back-up and standby generation.

Figure 2 shows the suggested timing of **capacity additions** and decarbonization upgrades (in MW) needed to fulfill the cumulative capacity progression in Figure 1 in support of data center demand growth.

Investment in grid electrification systems and software is needed to increase the capacity of existing transmission and to meet stringent data center power quality and reliability needs. For both grid-connected and behind-the-meter projects, the electricity demand profile needed to meet AI and data center load growth is unique and challenging. While the focus is on the size of the power demand

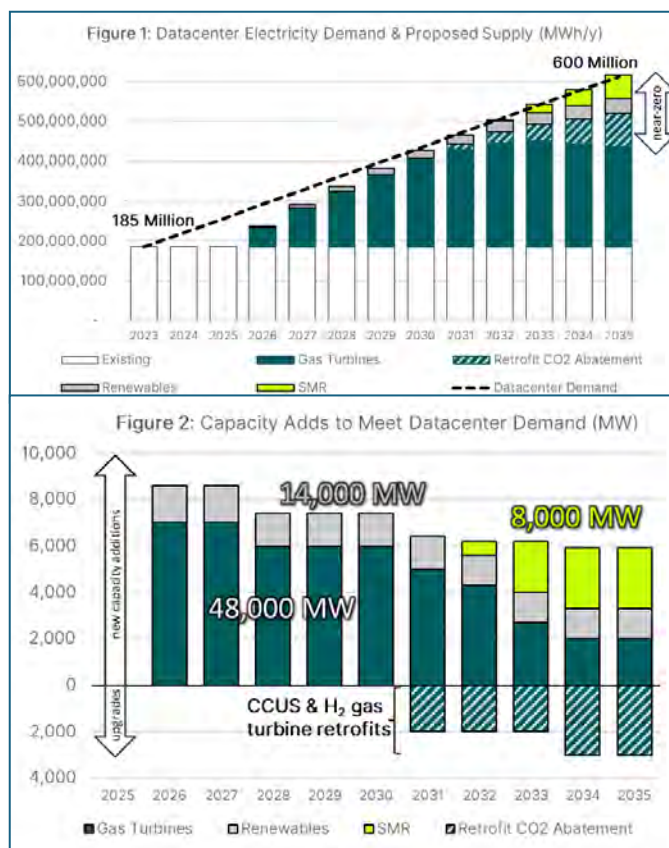
forthcoming, load shape is an equally important consideration. Not only are AI loads dense and significant, the millisecond fluctuations seen in training models requires intervention (load variability smoothing, stabilization systems, etc.) to ensure generation functions reliably. GE Vernova is developing and building such equipment and grid management software in the U.S.

The U.S. winning the data center and AI race is a key geopolitical and economic goal. Moreover, ensuring the energy and electricity feeding the data centers is provided by U.S. companies should also be seen as a geopolitical and economic imperative. Success in both efforts is possible.

III. SOLVING BOTTLENECKS TO ENERGY & ELECTRICITY DEPLOYMENT

The surge in electricity demand is also fundamentally reshaping grid dynamics. Data centers require gigawatt-scale power commitments, with low-latency, high-reliability energy supply. This is unique. Whereas traditional industrial load growth has evolved over decades, AI demand is materializing in just a few years, outpacing grid expansion timelines.

Areas of electricity pain points identified under the more traditional industrial load growth scenarios – such as interconnection bottlenecks, permitting delays, and the challenge of siting new generation and transmission infrastructure – are now exacerbated in the new AI and data center “space race” – threatening a scenario where traditional, segmented planning and approval processes do not advance quickly enough to keep pace.





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At the same time, AI workloads demand more than just megawatts—they require highly stable, resilient, and increasingly decarbonized power. This presents a dilemma: balancing grid reliability while integrating variable renewable energy, all while AI companies push for rapid deployment. New models are needed for capacity planning, flexible market structures, and behind-the-meter, and “grid-to-AI chip” solutions. Without strategic intervention, these challenges could lead to grid congestion, capacity shortages, and rising costs, making it clear that meeting AI-driven power demand will require both policy shifts and technology innovation.

At GE Vernova, we understand these dynamics and have the capabilities to partner with the administration in a unique and impactful way. Given our breadth, experience, and commitment to innovation, GE Vernova presents a powerful partnership opportunity for the Administration to not just diagnose the current gaps, but to help ensure they are filled effectively and expeditiously.

The section below dives deeper into four key bottlenecks: **dispatchable power, grid readiness, enabling systems, behind-the-meter needs, and innovation.**

a) **Dispatchable Power**

Hyperscalers have been reaching out to existing utilities to discuss the power needs they have. The responses from utilities have been consistent. There is insufficient existing generating capacity to meet their dispatchable power needs. Furthermore, through normal channels and processes, the timeline required to permit, bid, procure, build, and interconnect new generation to meet this demand will require several years longer than the hyperscalers can afford to wait.

As noted, AI data center electricity demand is unique and particularly challenging. The load shape is dense and significant and the millisecond fluctuations seen in training models requires additional consideration and intervention to help generation assets function reliably. Multi-GW data centers will require multiple blocks of power generation to meet compute power and thermal management needs 24/7/365. The solution to the constant electricity needs will likely be traditional on-grid electricity and services, behind-the-meter microgrid concepts, or a hybrid of the two. Regardless, the need for electricity to be continuous is certain.

An additional piece of this issue is **the need for additional dispatchable power to ensure the inertia and stability of the grid**. In essence, inertia is an essential property of the grid that ensures that the grid can resist changes in generation and demand while ensuring continuity of supply to load. Different generation sources produce different levels of inertia. Large rotating machines like gas turbines hold significant inertia. The lower the inertia on the system, the less stable a grid is. In this case, extensive use of high variability generation combined with the high reliability needs of AI can introduce challenges to enabling a stable power delivery inside data centers. Instead, it would be appropriate to emphasize the need for additional dispatchable power balanced with the installation of higher-variability energy sources that can help ensure an affordable, diversified electricity more broadly.

To reach sufficient dispatchable generating capacity for both industrial demand growth and the power needs for AI data centers, some recommendations follow:

1. **Ensure Upgrades to Existing Assets** – Incentive and streamline the permitting and regulatory process to allow for existing unit upgrades. For example, in North America



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technology upgrades can be added to just one class of our industrial gas turbines that will 14 GW to our system. Upgrading nuclear and restarting decommissioned sites can add 2 GW.

2. **Policy Supportive of Gas Generation Running at High-Capacity Factors** – The Trump Administration through its Executive Orders and other statements has been clear on the importance of gas power for American energy security and national security goals. It's important that policies and regulations align, including EPA rulemakings, permitting reform, and grid interconnects, at the earliest opportunity, with durability and certainty.
 3. **Expedited Deployment of Nuclear Energy** – Nuclear technologies that are ready for deployment should be an emphasis across the U.S. Government, including by ensuring the Nuclear Regulatory Commission (NRC) fulfills a proactive role in expediting new nuclear deployments. As GE Vernova's BWRX-300 is the only U.S. SMR design under development, such action would allow GE Vernova to expedite U.S. deployments. Domestically, we are partnered with the Tennessee Valley Authority, American Electric Power, and Duke Energy to advance U.S. development. Maintenance of the technology neutral credits (45Y/48E) which supports nuclear energy should also be maintained.
 4. **Cementing Support for New Gas Generating Capacity**– Given the need for significant capital investment across the supply chain for new gas capacity, Administration to Administration policy fluctuations regarding gas as a long-term electricity source has created uncertain investment cycles for original equipment manufacturers (OEMs) and suppliers. Spells of uncertainty can limit the deployment of capital required to build new capacity. Action to ensure clarity on the long-term viability of gas is recommended, including durable legal and legislative support to cement long term energy policies that benefit Americans and partners beyond political cycles, as well the innovation and deployment of technologies that reduce the emission intensity of gas over time.
 5. **Appropriate Consideration of Renewables** – With GE Vernova's unparalleled experience helping generate 25 percent of the world's electricity, we see an important role for renewable energy sources and a strong U.S. renewable energy supply chain. The inclusion of renewable energy allows for energy addition in a rapid manner. With a streamlined permitting process for renewable energy balanced with an intentional buildout of dispatchable energy, the U.S. can ensure a rapid response to electricity demand growth in a manner that diversifies the grid and reduces any operating cost risks from fuel price uncertainty or gas pipeline capacity. Beyond just the additional needed electrons on the grid, continued deployment of wind supports a U.S. wind supply chain that reaches 300,000 direct and indirect U.S. workers and creates skills and programs to lift up workers in a global era of competition.
 6. **Importance of Partnering with Full-Service Solutions Providers** – Given the complexity the grid and AI load, working with technology companies, like GE Vernova, that can provide the full suite of products and services necessary to enable a full-service system adds to the likelihood of success and efficiency in achieving a successful project.
- b) **Grid Readiness through Critical Grid Components and Software**



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The power grid is often referred to as the “most complex machine” ever built. With roughly 600,000 miles of high voltage transmission lines and 5.5 million miles of local distribution lines in the U.S., the network carries in excess of 4000 TWh of energy per year (the equivalent of 1.2 billion barrels of oil per year). It is the backbone of our economy, powering everything from the largest industrials to rural consumers. It is also foundational to hyperscalers as it will be counted on to get electrons consistently and effectively to them to power the country’s AI ambitions.

How the grid operates is often misunderstood. To perform reliably, the grid must be capable of matching supply and demand continuously. As demand rises, supply must ramp instantaneously. As demand falls, so must supply. It is a constant balancing act that continues to become more complex as generation and load increases. Specific to AI, the size and density of the loads combined with the frequent cycling of the loads (multiple times a second) can have unintended consequences on electricity voltage and oscillation if not managed with hardware and software. Unchecked, the voltage and oscillation fluctuations cause power supply problems that can lead to equipment malfunctions or damage impacting both the power generation equipment and the AI or data center equipment. The challenges are manageable, but it needs to be an area of focus.

The grid of today has not been designed to meet the emerging needs and current infrastructure requires modernization. To maintain the level of performance and reach required, it is estimated that for every \$1 invested in power generation, \$3 will need to be invested in grid infrastructure to facilitate effective and efficient electrons from the source of generation to consumption. As such, AI and industrial load growth isn’t just about more dispatchable generation but is also about the network that delivers these electrons.

The challenges with today’s grid are multiple. **Transmission bottlenecks are one issue.** Currently, the amount of long-distance transmission is inadequate for rapidly shifting demand centers, making it difficult to move power from generation sites to AI hubs. As a result, **regional networks are overloaded** as many key AI growth locations (Northern Virginia, Texas, the Pacific Northwest) already face severe congestion, limiting power availability. This issue is increased in magnitude due to the long approval timelines required to build new transmission lines – a process of up to 10-15 years due to permitting, land acquisition, and regulatory hurdles.

Interconnection delays also beleaguer efforts to attach new generation to the grid. 2,600+ GW of projects are waiting for grid access, with delays stretching 5+ years for approval. This backlog is not unique to any one technology. In many jurisdictions gas projects are delayed in a manner aligned to that of renewables. This is in part due to the **complex approval process**: studies and network upgrades are slow, costly, and often require coordinating multiple utilities and ISOs.

At the local level, **grid congestion and distribution constraints** are also emerging. Many prime data center locations are in regions with an already saturated grid that is currently at capacity, meaning local infrastructure upgrades are required before more load can be added. **Transformer and substation limitations** contribute to the localized issue as existing distribution infrastructure was not designed for the high density, high reliability loads AI demands.

Further, as load increases, the margins of excess electricity generation available is reduced to a range where new load connection will need to be halted to maintain the right levels of redundancy. This is driving hyperscalers to explore behind-the-meter (BTM) options. However,



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while this approach may work for hyperscalers, the BTM approach does not solve the grid issues facing the U.S. populace and is likely less efficient from a cost and reliability perspective.

As the grid continues to evolve with distributed generation, new flexibility, and new loads, its complexity increases and **real time visibility and control** becomes increasingly important for system operators. This dynamic is true across both the transmission system and the distribution system. Situational awareness and reactivity are no longer a luxury—it is essential to ensuring a reliable, resilient, and efficient grid in an era of unprecedented change. Beyond the hardware, more advanced automation and controls, like GE Vernova's Grid-Beats, and more contemporary software, like GE Vernova's GridOS, are needed to provide those who run our systems updated and accurate views as to what is happening across the grid and increase the ability to react.

Grid-Beats is a comprehensive portfolio of software-defined automation solutions aimed at streamlining grid digitalization and enhancing grid resilience through faster controls, artificial intelligence/machine learning (AI/ML)-based automation at the substation and transmission level. Alongside it, our interoperable GridOS software is built to orchestrate grid management by integrating energy data, network modeling, and AI-driven augmentation. **These systems are ones in which GE Vernova is also leveraging the benefits of AI.** By utilizing AI to process vast amounts of data in real-time we are enabling decentralized decision-making and proactive management of grid assets. With these capabilities, utilities can better optimize grid utilization, improve operational efficiency, unlock new economic opportunities, and bolster grid resilience.

Finally, **regulatory and market barriers** have a significant hand in today's grid issues. Grid expansion planning is not keeping pace with hyperscaler or market demand, leading to a growing mismatch between energy needs and infrastructure. Despite the need for more grid development, these regulatory and market barriers alongside decades of limited demand growth have combined to cause **a lack of demand certainty for manufacturers**. Due to the lack of investment into grid infrastructure in recent history, the necessity for U.S. manufacturing expansion of critical grid equipment has been limited. Today, **all power and grid equipment confront multi-year backlogs, with a limited U.S. supply chain for grid equipment**. Meeting the new need for electricity development and grid readiness requires ensuring manufacturing capacity to build and maintain some of the most critical equipment the grid relies upon—an essential factor in securing an abundant and reliable supply of energy to protect U.S. economic and national security

To overcome these barriers, there is an opportunity to both incentivize the grid as a priority and to declutter processes that have traditionally gotten in the way. Recommendations include:

1. Accelerated Transmission Expansion & Upgrades

- **Streamline Permitting** – Establish federal fast-track approval for high-priority transmission projects to cut approval timelines from 10-15 years to much less.
- **Expand Federal Siting Authority** – Allow the Federal Energy Regulatory Commission (FERC) to override state roadblocks for critical transmission corridors.
- **Incentivize Grid Investment** – Provide tax credits or financing support for transmission expansion near high-demand AI hubs.



2. Reformed Interconnection Processes

- **Prioritize Ready Projects and Firm Capacity** – Prioritize projects that are "shovel-ready" to reduce speculative queue congestion, fast track firm dispatchable generation.
- **Mandate Faster Studies and Cost Transparency** – Require ISOs and utilities to complete interconnection studies within 12-18 months.
- **Preemptive Grid Reinforcements** – Encourage grid operators to proactively upgrade infrastructure in AI growth regions rather than waiting for interconnection requests.

3. Upgrading Local Grid Infrastructure

- **Modernize Distribution Networks** – Increase federal and state funding for substation, transformer, and voltage control upgrades in key AI data center regions.
- **Encourage Private Investment in Grid Capacity** – Allow hyperscalers to co-fund or co-develop local transmission and distribution upgrades with utilities.
- **Develop Microgrid & Hybrid Models** – Support AI hubs in building integrated grid-connected microgrids that reduce reliance on centralized power delivery.

4. Aligning Energy Markets with Hyperscaler Demand

- **Flexible Rate Structures** – Adjust electricity pricing models to incentivize AI-driven high-load customers to locate in grid-friendly areas.
- **Demand-Side Grid Participation** – Enable hyperscalers to contribute to grid stability via energy storage, demand response, and backup generation services.

5. Expanding the Availability of Critical Equipment and Create Strategic Reserve

- **Manufacturing Support** - Expand funding for domestic manufacturing expansion of critical grid equipment and components at U.S.-based factories, including through the Defense Production Act Article III provisions.
- **Strategic Reserves** - Establish a strategic reserve of critical grid equipment.

6. Improved Treatment of Grid Reliability Measures

- **Flexibility in Grid Software Upgrades** – Direct utilities and utility commissions to allow software upgrades to be funded by changes to the utility rate base in order to maintain up-to-date safety, cybersecurity, and reliability programs.
- **Security Standards Alignment** – Unify regulatory requirements for grid reliability, cybersecurity, and AI/autonomous control standards.

GE Vernova is ready and willing to help in providing the technologies that will be required to build out this infrastructure. GE Vernova can also provide horizontal capabilities within our Consulting Services unit that brings to the forefront the capabilities needed to prioritize and deploy technology in the most effective and efficient ways possible.

c) Enabling Systems & Behind-the-Meter Approaches



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The need for expected build out of infrastructure to meet both the time horizons and capability needs of hyperscalers necessitates that all options are on the table. Given many of the challenges noted across the network today (interconnection queues, permitting timelines and complexity, and sensitivity around impact to consumer rates), hyperscalers are assessing alternative and non-traditional approaches to get the power they need when they need it.

One non-traditional approach emerging is behind-the-meter (BTM) projects. BTM approaches are developing due to the absence of available grid-connected generation and transmission. At this time, **the most prevalent options for BTM applications are either islanded and completely isolated from the bulk electric grid or connect “within the fence” of an existing grid-connected power plant.** Islanded configurations allow for quicker deployment of generation than new grid connected generation due to the interconnection process, but most BTM applications will require back-up redundancy of turbines, energy storage and stability solutions, coordinated with multiple levels of electrical controls to keep the electric system stable and reliable. Additionally, BTM applications that are created from transitioning existing grid connected generation to dedicated power for a specific data center consumes a substantial amount of power that would normally be delivered to electric consumers, creating a resource adequacy gap. Models may emerge in which a BTM application takes a hybrid approach, developing new generation capacity that serves the data center while selling excess to the grid.

These BTM structural models should be considered when assessing the nations desired approach. Planning should recognize that **BTM offers benefits (such as speed to deployment), while noting that BTM approaches must be balanced with the opportunity to make the overall power system more reliable and efficient** for AI and data center electricity needs and long-term industrial growth and affordability. A framework that balances speed, power quality, impact on the broader system, and economic feasibility must exist and be leveraged intelligently.

Additionally, within both BTM and traditional grid approaches, there exists a unique need for more pre-integrated solutions if we expect to move at the pace necessary as an industry. It’s not just about the generation technologies; but also, the grid components (substations, transformers, switchgears) that integrate seamlessly to deliver power outcomes that hyperscalers need. Grid software must be utilized to tie all components together and provide visibility to the “system” while enabling rudimentary controls that adjust in near real time to the normal fluctuations in power delivery and transmission will be essential. Integrating these solutions requires understanding the increased complexity of both on-grid and off-grid power for hyperscalers.

Achieving pre-integrated solutions requires a “grid-to-AI chip” approach with novel architectures based on Medium Voltage Direct Current (DC) and power stabilization for increased efficiency, stability, and reliability. Such new architectures require innovation such as solid-state transformers, DC protection systems, medium voltage uninterruptible power supply (UPS), and battery storage combined with advanced energy management systems. This is a space where GE Vernova is leading in developing and deploying new solutions.

Critical to the BTM option is **reliability and resiliency risk**. As noted, hyperscalers require ultra-high uptime, yet off-grid solutions lack the grid’s inherent redundancy. This places greater pressure on private operators to ensure self-sufficient, always-on power without the grid’s stabilizing effect. Further, many energy policies and market structures are designed around grid-



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connected customers. Off-grid hyperscaler projects may operate in regulatory gray areas, lacking access to market mechanisms like capacity markets or demand response programs. Additionally, large-scale, self-sufficient power generation requires significant land and fuel access, often straining local resources and regulatory frameworks - **reinforcing the need for streamlined permitting approvals**. Permitting and operational challenges also arise when hyperscalers demand immediate electricity access and turn to off-grid power using fossil fuels; yet also require long-term pathways to decarbonization of these assets. This fragmented energy planning risks inefficiencies, as private generation may not align with regional grid expansion strategies.

Grid planners must factor in **national and regional energy security**. As hyperscalers move BTM, policy must consider the grid stability, market competitiveness, and energy resource allocation implications. Interoperability between private and public assets could become critical.

The following recommendations would help in increasing speed and efficiency of build outs:

1. Streamlined Permitting & Siting

- **Expedited Permitting for both Private and Utility Generation Serving Hyperscaler Demand** – Create fast-track approval processes for hyperscaler-led energy projects, including natural gas, renewables, and battery storage.
- **Standardized Siting Rules** – Establish clear, uniform regulations across jurisdictions to reduce permitting bottlenecks and delays.

2. Energy Market & Interconnection Reform

- **Flexible Grid Participation Models** – Allow off-grid systems to engage in limited grid services (e.g., capacity markets, emergency backup) without full interconnection burdens.
- **Interconnection Fast-Track for Hybrid Solutions** – Prioritize interconnection approvals for projects that blend on-site generation with partial grid access.

3. Fuel Supply & Infrastructure Support

- **Enhanced Natural Gas & Hydrogen Access** – Streamline regulatory approvals for pipeline and storage infrastructure to ensure reliable fuel supply for off-grid generation.
- **Incentives for On-Site Energy Storage** – Provide tax credits or grants for large-scale battery deployment to enhance reliability without full grid dependence.

4. Tax & Investment Incentives

- **Expanded Investment Tax Credit (ITC) & Production Tax Credit (PTC) Eligibility** – Extend existing tax credits to include private hyperscaler energy projects inclusive of gas.
- **‘Speed to Power’ Incentives** – Implement grants or funding mechanisms that reward projects delivering rapid, reliable energy solutions for AI and data centers.

It is likely that the ultimate solve will be a combination of “Grid Connected” and “Off Grid”. In both instances, GE Vernova has the capability to support as evidenced by our work with



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customers engaging in both scenarios. Beyond comprehensive equipment and services portfolio across the generation and electrification landscapes, our in-house expertise can provide needed support in the areas of system solution stability, resiliency, and efficiency in all scenarios.

d) Spurring American Energy Sector Innovation While Powering Data Centers

The opportunity to power data centers is not only about enabling power today. Investments made at scale will pay other significant dividends, including driving innovation. This required innovation is a positive. While the primary focus now is meeting the energy demand required to lead in AI and data centers, global energy technology deployment is itself an important geopolitical and economic race. Countries who lead in innovation such as SMRs, carbon capture, and hydrogen, and whose technologies and energy resources are deployed globally, will benefit from domestic job growth, increased manufacturing, and improved geopolitical standing born from the international partnerships associated with meeting global energy and electricity needs.

However, competition in the domestic and global energy ecosystem requires continuous innovation to meet the needs and demands of customers. To ensure the U.S. remains leaders of global competition, U.S. innovators and technology manufacturers must achieve cost reductions in manufacturing, increase the efficiency of established technologies, and bring new breakthrough technologies demanded by customers to market to affordably in order to solve energy poverty while meeting rising electricity demand. It is vital that the U.S. Government builds strong partnerships with companies like GE Vernova to drive this innovation.

One such example: many hyperscalers and utility customers have long term emission profile objectives and even state regulatory requirements to address their carbon footprint from operations. The opportunity to meet the full demands of electricity while innovating and deploying lower carbon intensity technologies at scale can further the trajectory toward such goals. For example, GE Vernova in 2023 added 29 GW of capacity to the global grid while energizing 64GW of transformers. These energy additions were at a carbon intensity 25 percent below the grid as a whole, thus improving the trajectory of carbon emissions. Beyond improving carbon intensity, innovating breakthrough technologies such as the potential to retrofit gas assets built today with carbon capture technologies or to operate these assets with hydrogen in the future helps achieve the goals of the hyperscalers and advance U.S. leadership in innovation.

We believe the American spirit of innovation cannot be told without us. With a history of innovation that traces back to Thomas Edison, generations of Americans can point to family members who worked with GE Vernova to build the manufacturing base and electricity system that helps power 25% of the world's electricity. The innovation ethos carries through today as we invest \$1+ billion per year into breakthrough technologies. Our research is driven by the needs of our customers domestically and abroad – ensuring the work we do aligns with market need.

Our research is wide-ranging. It includes the BWRX-300 – the cutting-edge innovation from Wilmington, NC, providing a smaller and simplified version of 80+ deployed GE Vernova nuclear reactors. Our gas and fuels teams are delivering the first gas-fired power station with carbon capture and storage as part of the Teesside Project in the United Kingdom; as well as, developing new carbon capture and direct air capture solvents while driving efficiency improvement and hydrogen integration in our gas turbine offers. Similarly, we are leading in



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electricity equipment and systems research, including medium voltage DC systems with solid state transformer protection systems, DC uninterruptible power supply (UPS) advanced energy management systems, and electrification systems for high-heat industrial applications.

As highlighted previously, **GE Vernova is proud to be further integrating machine learning and AI into our power system control software to drive a better performing, more affordable grid.** By utilizing our Grid Beats offering to create centralized virtual protection and control systems for substations and transmission applications and GridOS to manage the grid, these software systems enable decentralized zonal autonomous control – an approach that divides the grid network into autonomous zones that function efficiently together but can maintain relative independence to operate independently in case of disruption.

To help ensure future U.S. innovations that will support the AI and data center race, we propose:

1. **Research Partnerships** - Ensure Government funded research is done in partnership and aligned with U.S. private sector innovators who will commercialize innovation.
2. **SMR Deployment** - Enable U.S. SMR development through existing programs at the Department of Energy and potential early build cost-override insurance.
3. **Early-Stage Research** - Continue early-stage R&D programs like the Advanced Research Projects Agency – Energy (ARPA-E).
4. **Competitive Advantage Development** - Support DOE programs where the U.S. can lead, including carbon capture, electrification hardware, power systems controls, distributed generation integration, and industrial electrification.
5. **Early Deployments** - Guarantee financial and regulatory support for early deployments of breakthrough U.S. technologies to ensure first deployments in the U.S. are successful while developing supply chains and Intellectual Property (IP).
6. **Supply Chain and Labor** - Solve domestic supply chain and labor constraints to ensure the prevention of future bottlenecks that could limit global market opportunity.
7. **Department of Defense** - Expand the Department of Defense's R&D to include energy security, grid hardening, and microgrids that serve the front line as well as the U.S. market.

IV. CONCLUSION

The Administration is right to prioritize the AI and data center race, as well as the need for growth in generation and grid development in the United States. The U.S. addressing the global data center and AI competition is a key geopolitical and economic goal. Within this, ensuring the energy and electricity feeding the data centers is provided with U.S. technology should also be seen as a geopolitical and economic imperative. Success in both efforts is possible.

GE Vernova is aligned in this mission. As the American energy manufacturer serving the full spectrum of generation and grid, we are well positioned to partner with policymakers and customers and to help design and build solutions required to meet growing data center electricity demand while making energy more secure and affordable for everyone. We look forward on the opportunity to engage further with the Administration and our partners together on these topics.